

Fig. 3: Actual-size, single-side PCB pattern of the 0-50V variable power supply

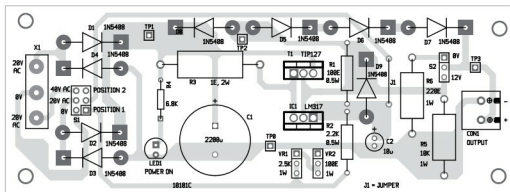


Fig. 4: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - LM317 adjustable voltage regulator
 T1 - TIP127/2N6107 pnp transistor
 D1-D9 - 1N5408 rectifier diode
 LED1 - 5mm LED
- Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):
 R1 - 100-ohm, 0.5W
 R2 - 2.2-kilo-ohm, 0.5W
 R3 - 1-ohm, 2W
 R4 - 6.8-kilo-ohm
 R5 - 10-kilo-ohm, 1W
 R6 - 220-ohm, 1W
 VR1 - 2.5-kilo-ohm, 1W potentiometer
 VR2 - 100-ohm, 1W potentiometer
- Capacitors:
 C1 - 2200 μ F, 75V electrolytic
 C2 - 10 μ F, 75V electrolytic
- Miscellaneous:
 S1 - DPDT switch
 S2 - SPDT switch
 S3 - On/off switch
 CON1 - 2-pin connector terminal
 X1 - 230V AC primary to 20V-0-20V, 3A secondary transformer
 - Heat-sinks for LM317 and TIP127

So, if LM317 is fed with a maximum of 40V, output will be 40V-3V = 37V

Circuit diagram of the 0-50V, 1A variable power supply is shown in Fig. 2. It is built around step-down transformer X1, bridge rectifier comprising D1 through D4 (1N5408), adjustable voltage regulator LM317 (IC1), power transistor TIP127 or 2N6107 (T1) and diodes 1N5408 (D5 through D9).

The circuit is centred around LM317, whose output depends on resistors R1, R2, VR1 and VR2. A centre-tapped transformer (X1) with 230V AC primary to 20V-0-20V, 3A secondary is chosen, whose centre-tap point is at 0V and voltage between both ends of secondary windings is 40V AC. So, voltage between the centre tap and either end of the secondary winding is 20V AC. One side (half) of the secondary (20V AC) windings is connected to anode of D1 and cathode of D4.

Secondary of X1 is connected to the bridge rectifier through range se-

lector switch S1. When S1 is in position 1, its pins 3 and 1 as well as pins 4 and 2 are shorted. This results in connecting 20V AC to the bridge rectifier. The 20V AC is rectified, producing around 29V DC, which is connected as input to LM317 IC.

Resistor R2 is shorted through pins 1 and 3 of S1, connecting VR1 and VR2 at pin 1 of IC1. Output depends on the positions of VR1 and VR2. VR1 is configured as a coarse voltage-adjusting knob and VR2 is configured as fine voltage-adjusting knob. By varying the potentiometers (VR1 and VR2), voltage output is varied in the range of 1.2V to 27.8V (at TP2).

With S1 in position 2, its pins 1 and 3 are opened and pins 4 and 6 are shorted. This results in connecting 40V AC to the bridge rectifier. Thus 40V AC is rectified, producing around 56.4V DC, which is connected as input to LM317. Shorting of R2 is now opened. Due to a series connection of R2 with VR1 and VR2, lower limit of the output voltage is increased from 1.2V to 27V. At position 2 of S1, differential voltage of LM317 is 56V-27V = 29V. By varying potentiometers VR1 and VR2, in position 2 of S1, voltage is varied in the range of 27V to 50V at TP2.

At output stage, three diodes (D5 through D7) are connected in series to obtain 0V output, assuming each diode voltage drop of 0.5V (that is, $3 \times 0.5 = 1.5$ V) nullifies output residue voltage of 1.2V from LM317. Zero-volt selector switch S2 works in 0V or 1.2V mode. When it is in 0V mode, effect of diodes in series is present and so 0V output is obtained. When the switch is in 1.2V mode,